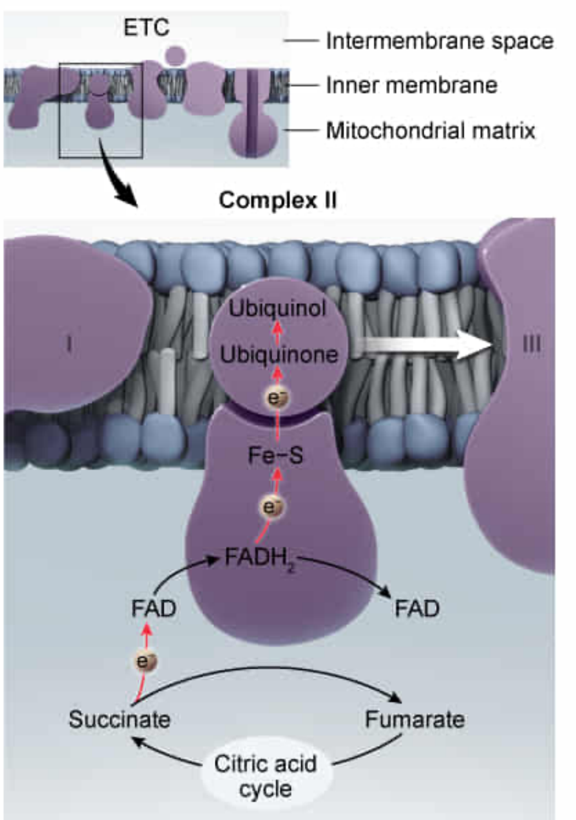
Atpenins are succinate-ubiquinone reductase inhibitors with antifungal properties. Atpenins illustrate the fact that:

- A. succinate oxidation at Complex II decreases the pH of the intermembrane space. [19%]
- B. protons from $FADH_2$ flow back into the matrix against their concentration gradient. [7%]
- ✔ C. electron transfer from Complex II to III is facilitated by ubiquinol in the inner membrane. [59%]
- D. ubiquinone has a higher reduction potential than cytochrome *c* in the cell membrane. [13%]

Correct
59% answered correctly

Explanation:



ETC = Electron transport chain ©UWorld

Succinate-ubiquinone reductase, also known as Complex II or succinate dehydrogenase, is an oxidoreductase found in the inner membrane of the mitochondria in eukaryotes. In the electron transport chain, Complex II facilitates the transport of electrons from **succinate** (in the citric acid cycle) to **Complex III**. Electrons move through Complex II by three consecutive reactions. First, electrons from succinate are transferred to flavin adenine dinucleotide (FAD) to produce $FADH_2$ as succinate is oxidized to fumarate. $FADH_2$ is then reoxidized to FAD when electrons are shuttled to the iron-sulfur centers in Complex II. Finally, a small mobile electron carrier called **ubiquinone**, or coenzyme Q, accepts electrons from the iron-sulfur centers, becoming reduced to **ubiquinol**. Ubiquinol then transports its electrons to Complex III.

Electrons transferred by Complex II contribute to the overall production of ATP, which provides cells with the energy to function. The question indicates that atpenins are effective antifungal agents that inhibit succinate-ubiquinone reductase (ie, Complex II). Consequently, the correct choice should illustrate that atpenins disrupt at least one of the three reactions of Complex II. The transfer of electrons from Complex II to III through ubiquinol is the only reaction accurately described.

(Choices A and B) Although Complex I, III, and IV pump protons into the intermembrane space against their concentration gradient, Complex II does *not* pump protons, and therefore does *not* directly contribute to the decrease in pH in the intermembrane space.

(Choice D) In the electron transport chain, carriers with a higher reduction potential have a stronger tendency to be reduced (gain electrons). Therefore, electrons are transferred to carriers with higher reduction potentials at each step. Because electrons are transferred *from* ubiquinol *to* cytochrome *c*, this indicates that ubiquinone has a *lower* reduction potential whereas cytochrome *c* has a higher reduction potential.

Educational objective:
Complex II (succinate-ubiquinone reductase, or succinate dehydrogenase) oxidizes succinate to fumarate and transfers electrons to Complex III. Electrons are transferred from flavin adenine dinucleotide ($FADH_2$) to iron-sulfur centers in Complex II, then to ubiquinone (coenzyme Q), which carries them to Complex III through the inner membrane in the form of ubiquinol.

Time spent:0 sec
Subject:
Biochemistry

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System:
Metabolic Reactions